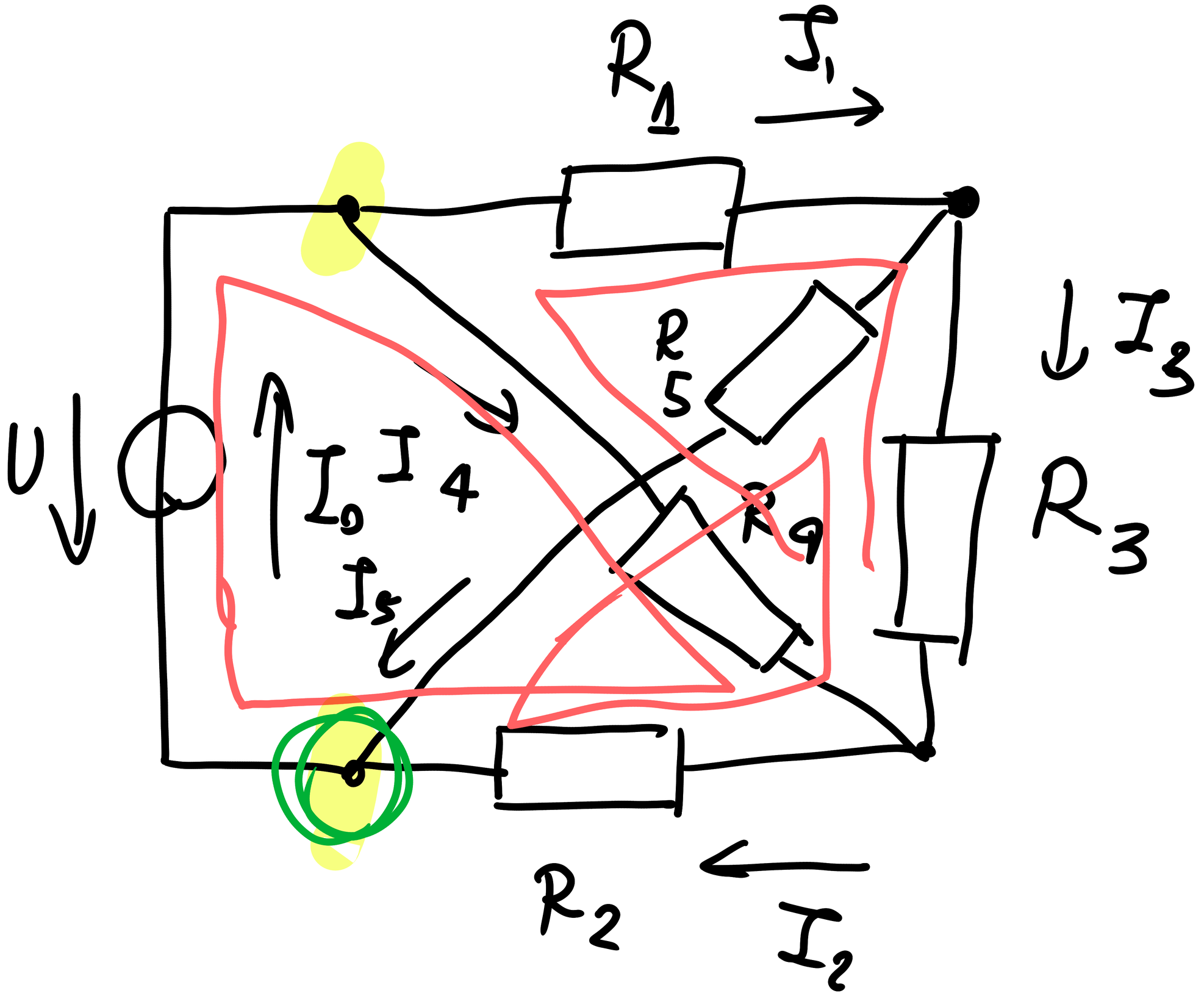
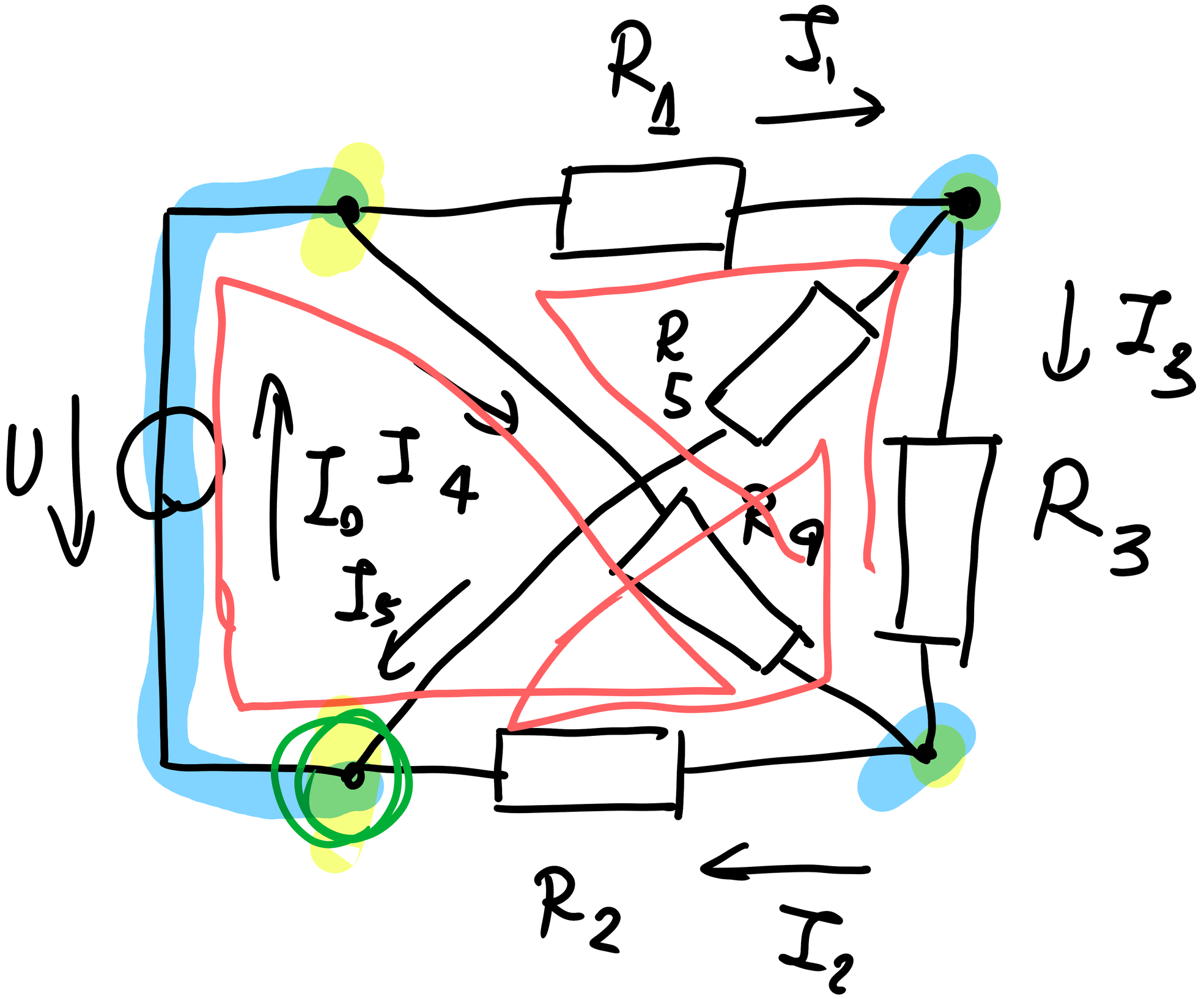
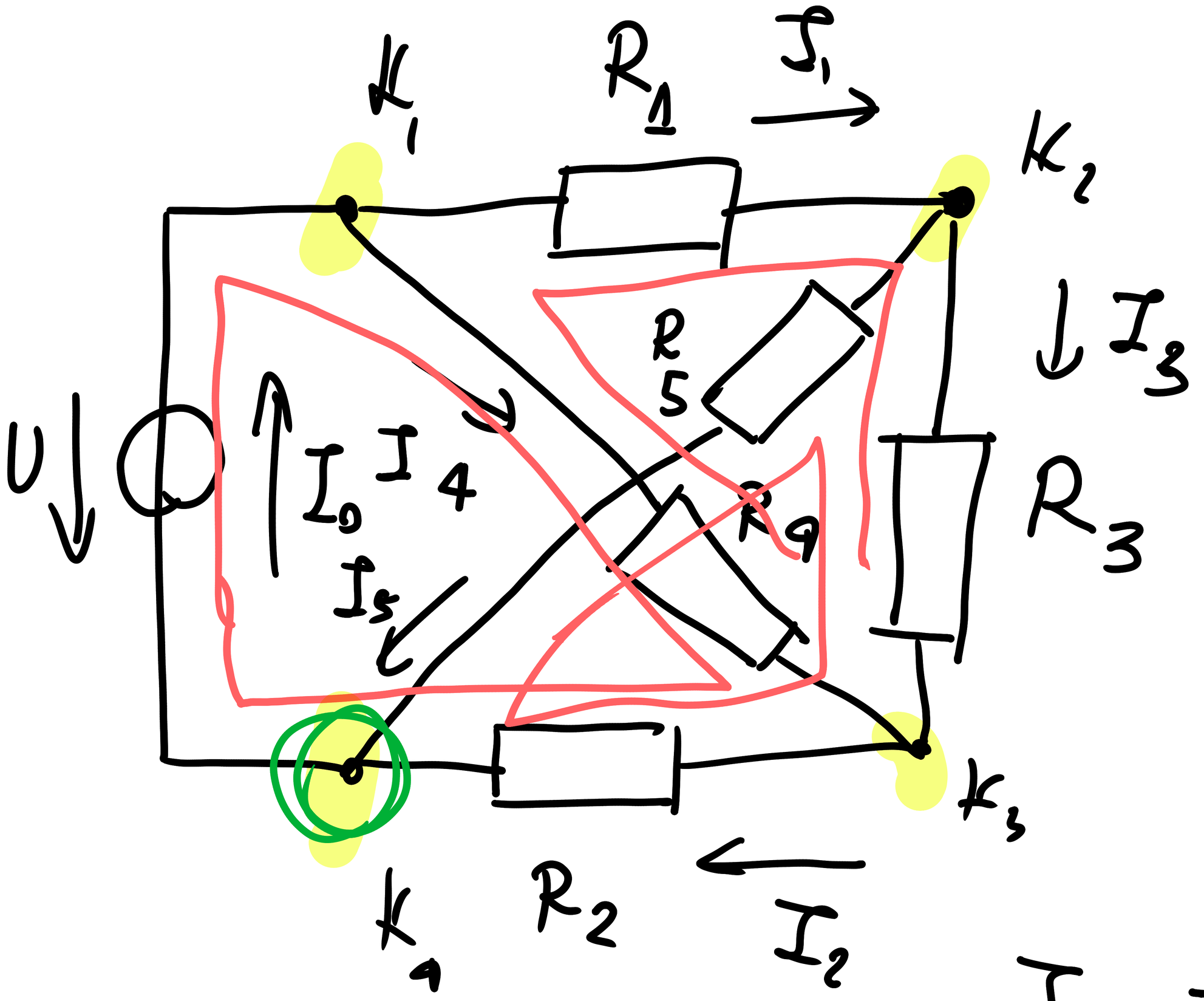


$$I = I_3 + I_1$$

$$I_0 - I_5 - I_2 =$$


R1
J1
R5
J3
U
I0
I4
R4
R3
I5
R2
I2



$$I_0 - I_5$$

$N = 6$

$$\left.\begin{array}{l} K_1: I_0 - I_1 - I_4 = 0 \\ K_2: I_1 - I_3 - I_5 = 0 \\ K_3\ I_3 + I_4 - I_2 = 0 \end{array}\right\} \quad K - 1 = 3$$

$$K_4: I_0 - I_5 - I_2 = 0$$

$$I_0 = I_2 + I_5$$

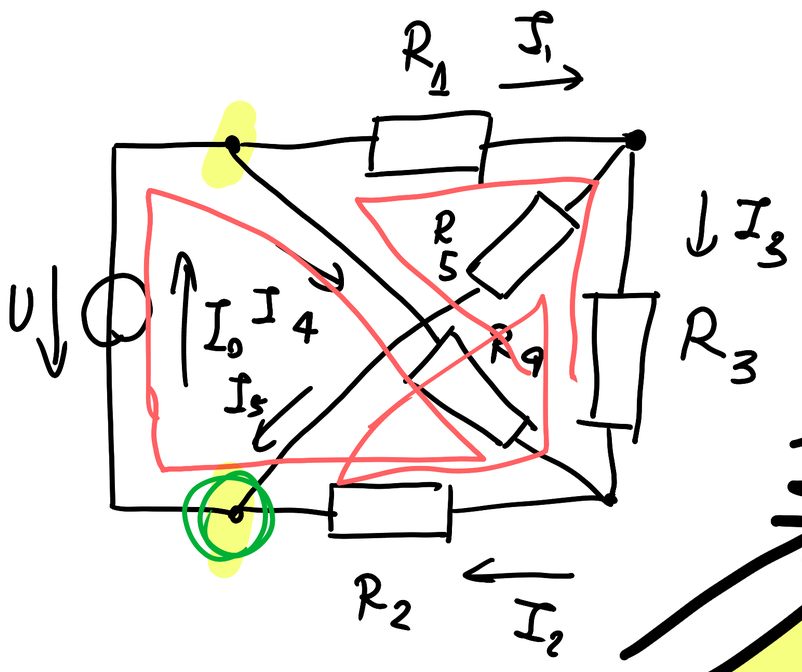
$$K_1: \; I_2 + I_5 - I_1 - I_4 = 0$$

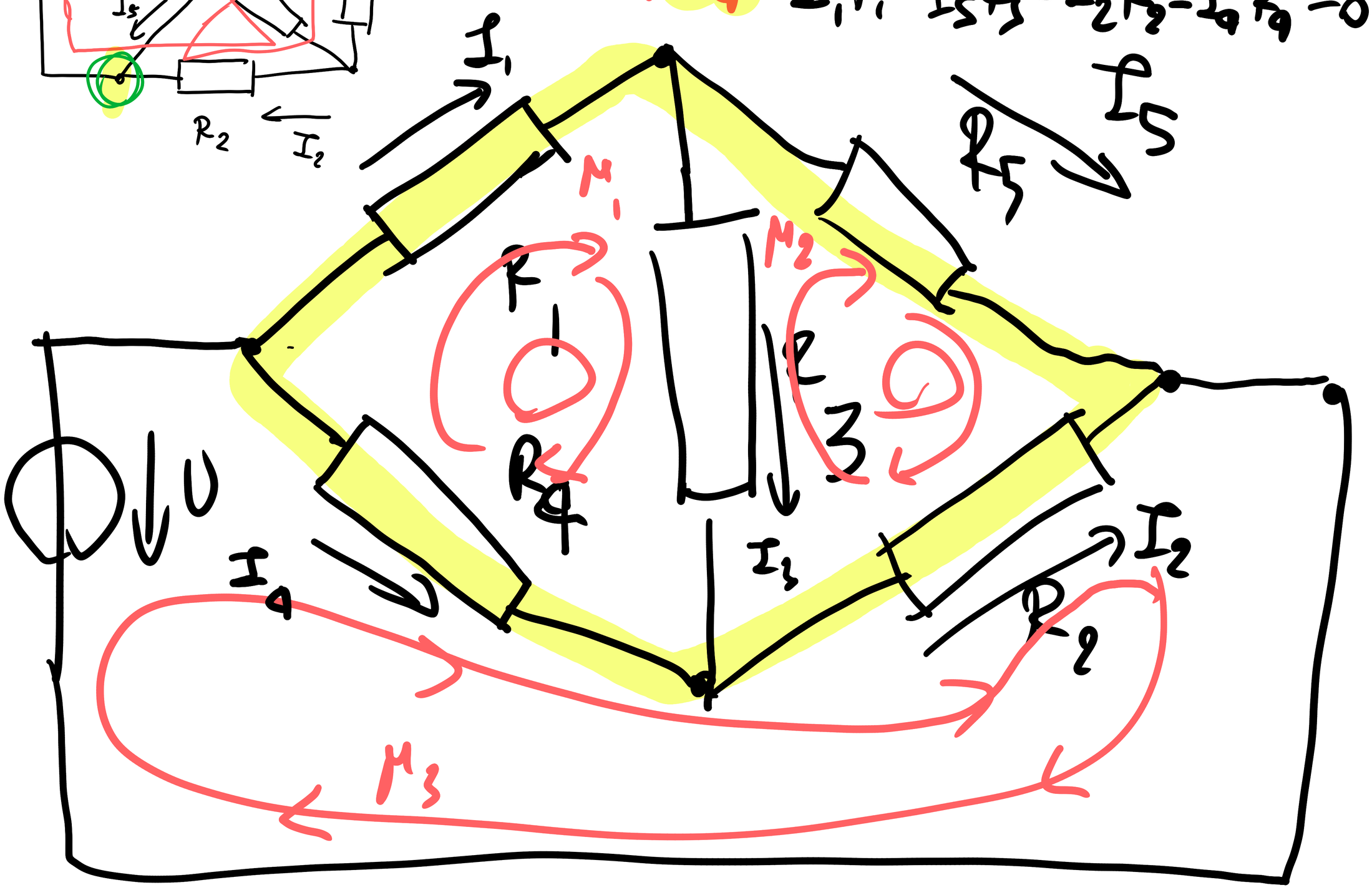
$$M_1: I_1R_1 + I_3R_3 - I_4R_4 = 0$$

$$M_2 \quad I_5R_5 - I_2R_2 - I_3R_3 = 0$$

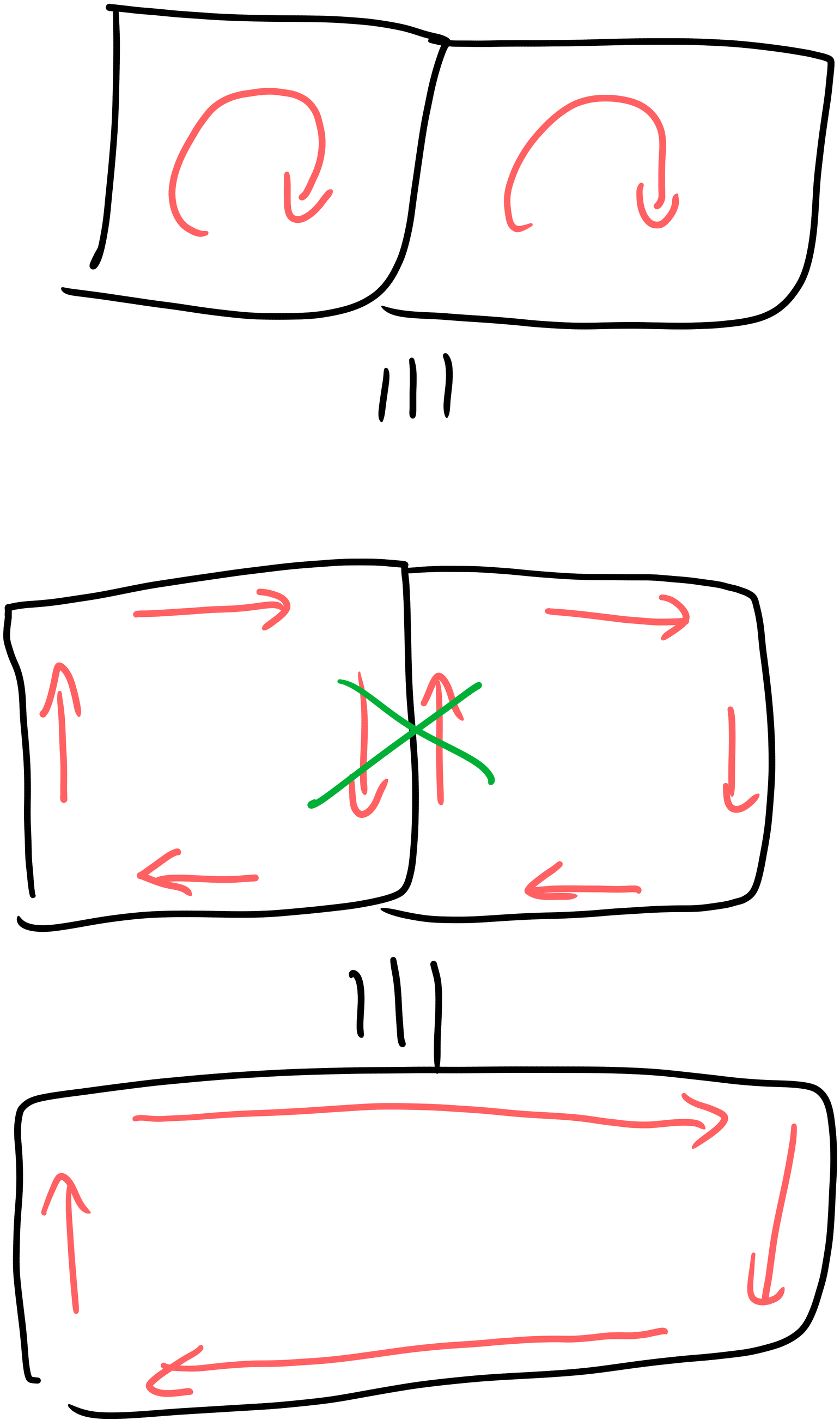
$$M_3 \quad I_4R_4 + I_2R_2 - U = 0$$

$$M_4 \quad I_1R_1 + I_5R_5 - I_2R_2 - I_4R_4 = 0$$



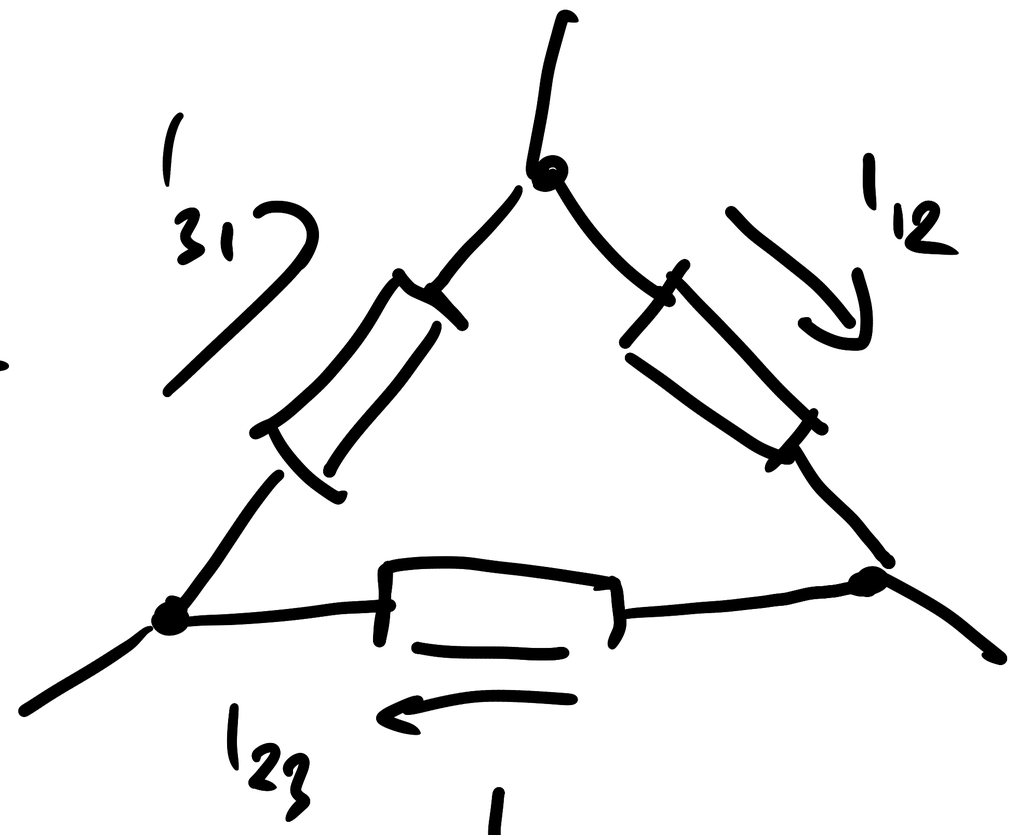


$$M_4 = M_1 + M_2$$

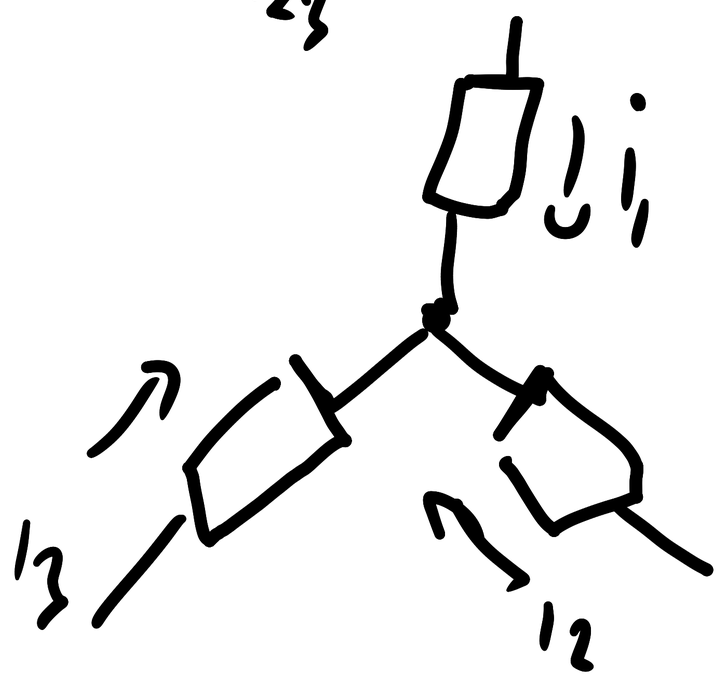


$$\begin{bmatrix} 1 & -1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 1 & 1 & 0 \\ 0 & R_1 & 0 & R_3 & -R_4 & 0 \\ 0 & 0 & -R_2 & -R_3 & 0 & R_5 \\ 0 & 0 & R_2 & 0 & R_6 & 0 \end{bmatrix} \begin{bmatrix} I_0 \\ I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ U \end{bmatrix}$$

$$R_1 = \frac{R_{12} \cdot R_{31}}{R_{12} + R_{23} + R_{31}}$$



$$R_{12} = R_1 + R_2 + \frac{R_1 R_2}{R_3}$$

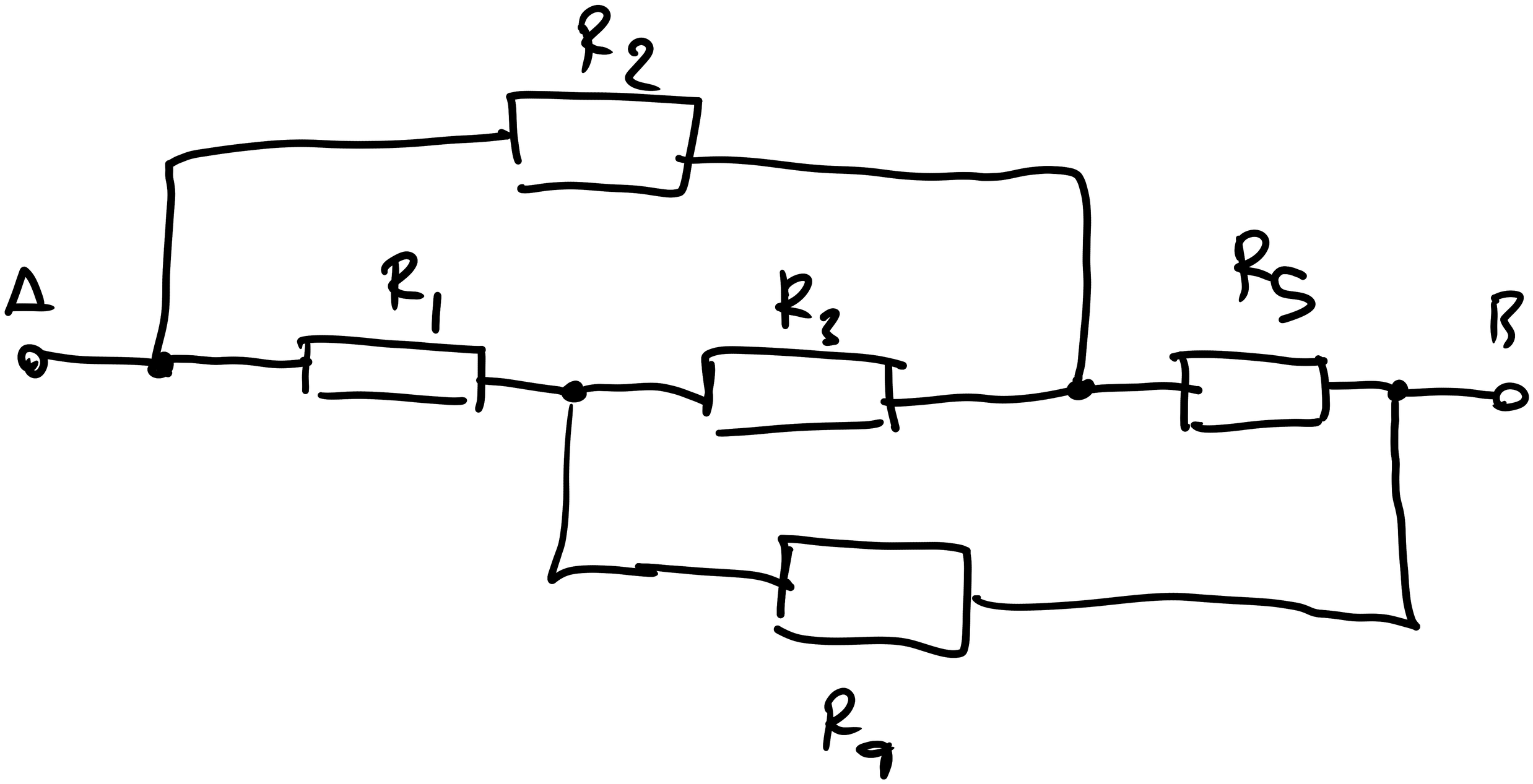


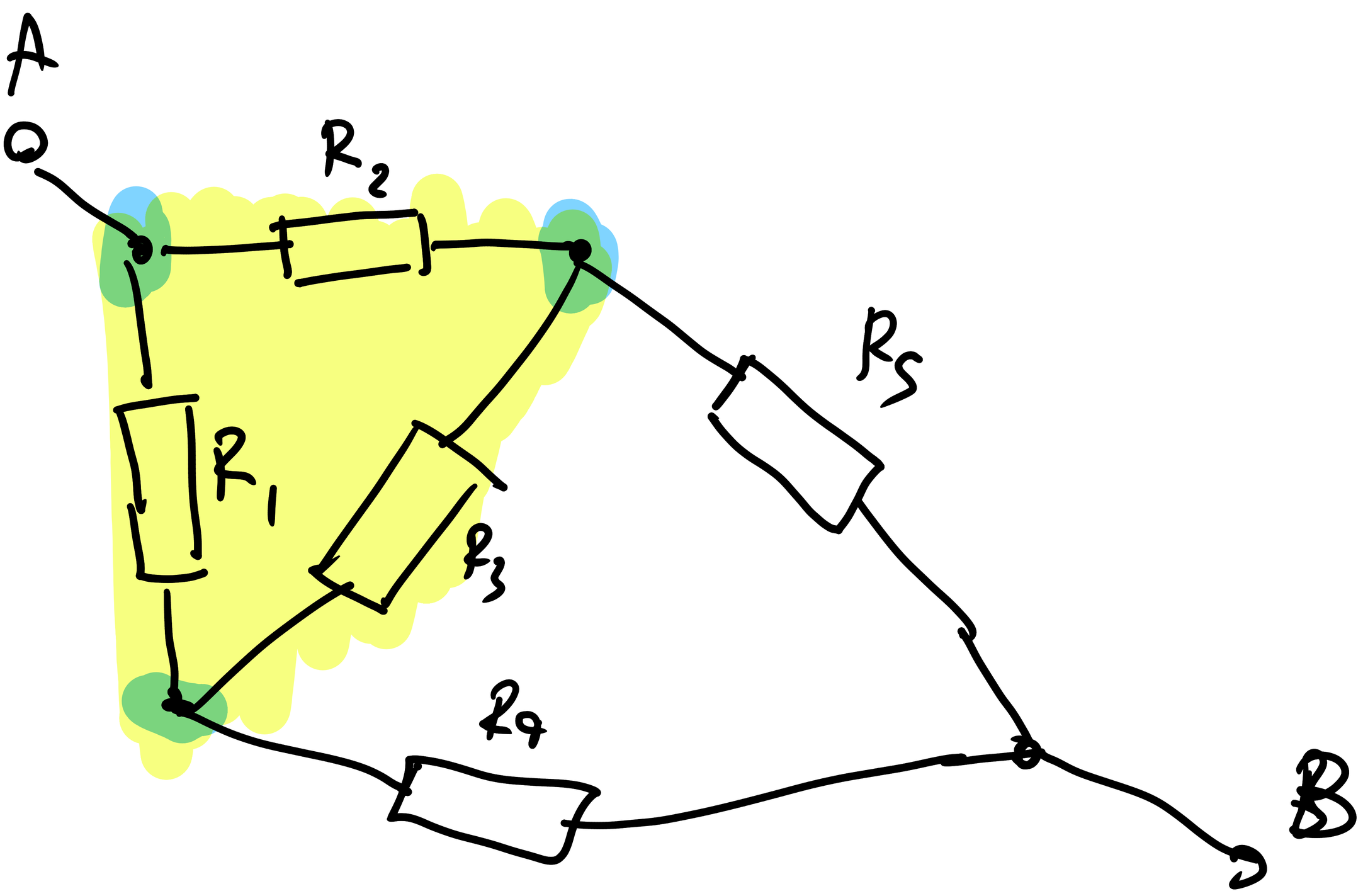
$$I_{12} - I_{31} = i_1$$

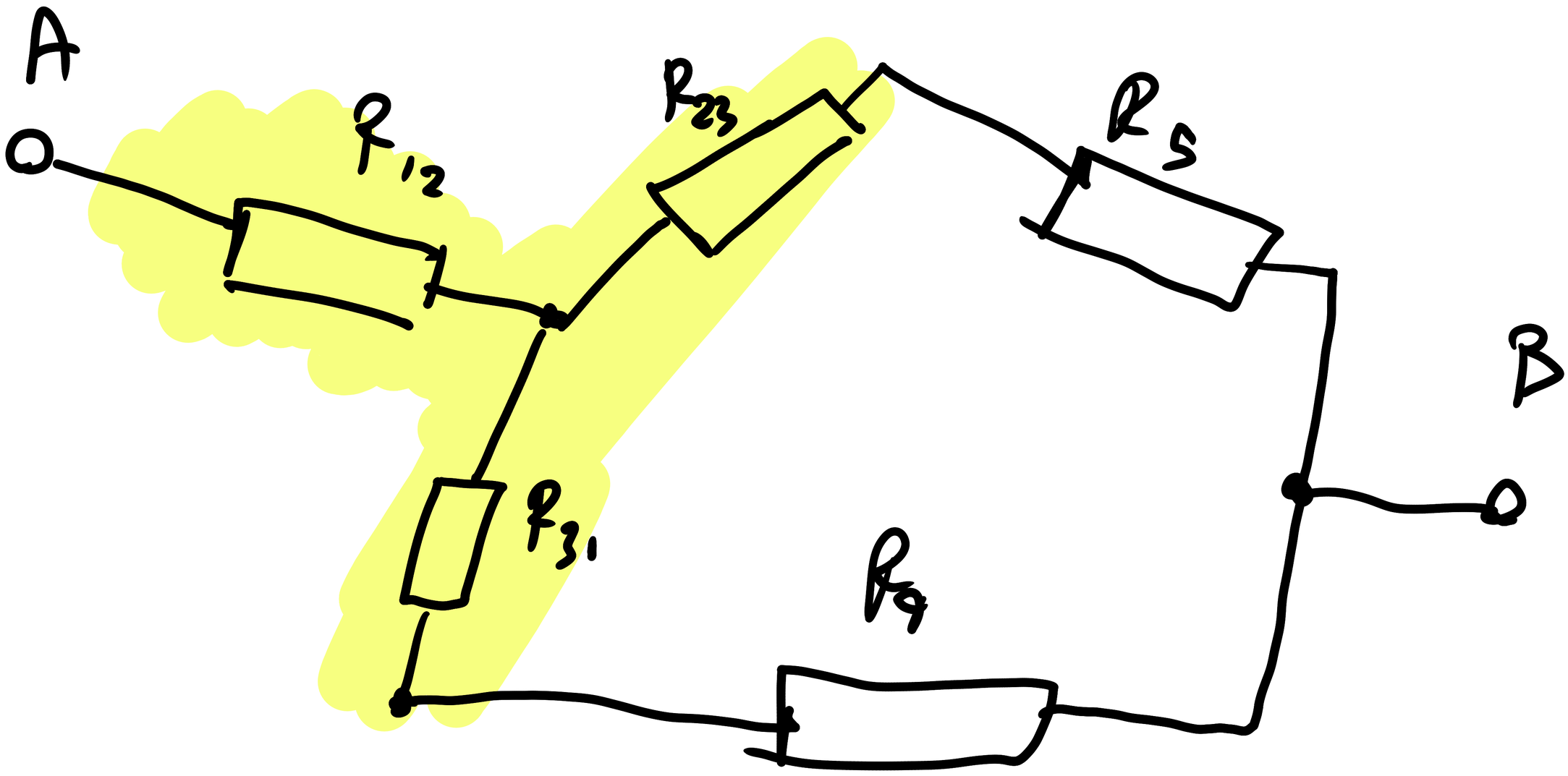
$$R_* = \frac{R_\Delta \cdot R_\Delta}{3R_\Delta} = \frac{1}{3} R_\Delta \Rightarrow R$$

$$R_1 = R_2 = R_3 = R_* = R_Y$$

$$R_{12} = R_{23} = R_{31} = R_\Delta$$

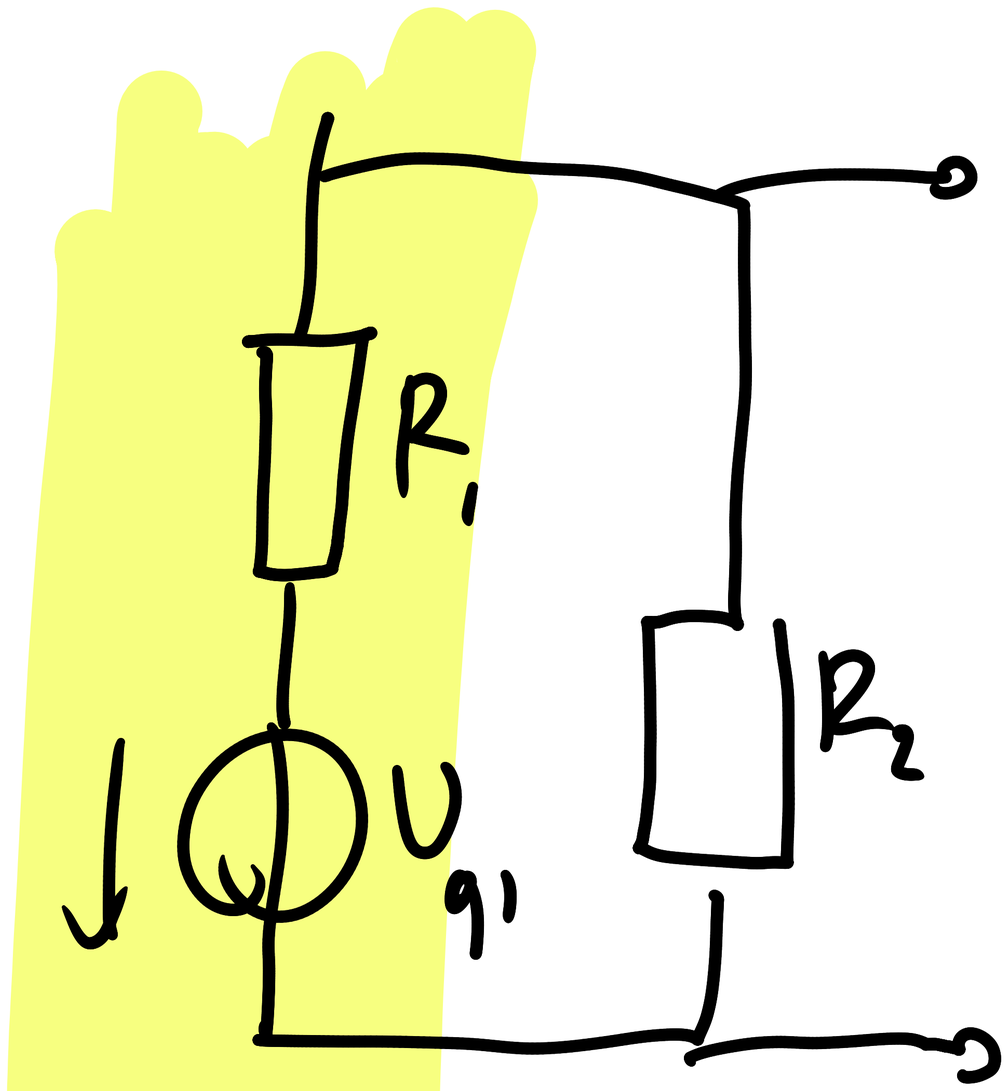

A
R2
R1
R3
R5
B
R4


A
R2
R1
R3
R5
R4
B

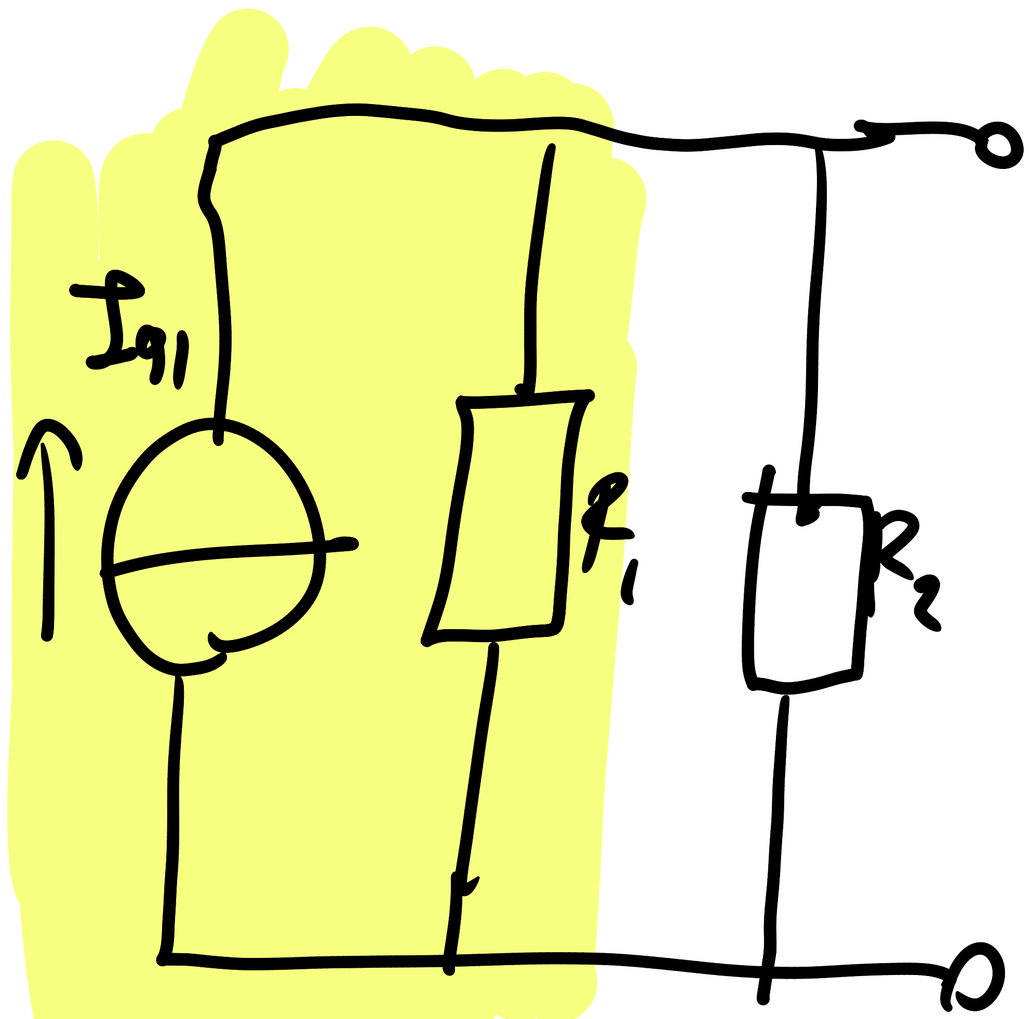


$$R_{AB} = R_{12} + \frac{(R_{23}+R_5)(R_{31}+R_4)}{R_{23}+R_5+R_{31}+R_4}$$

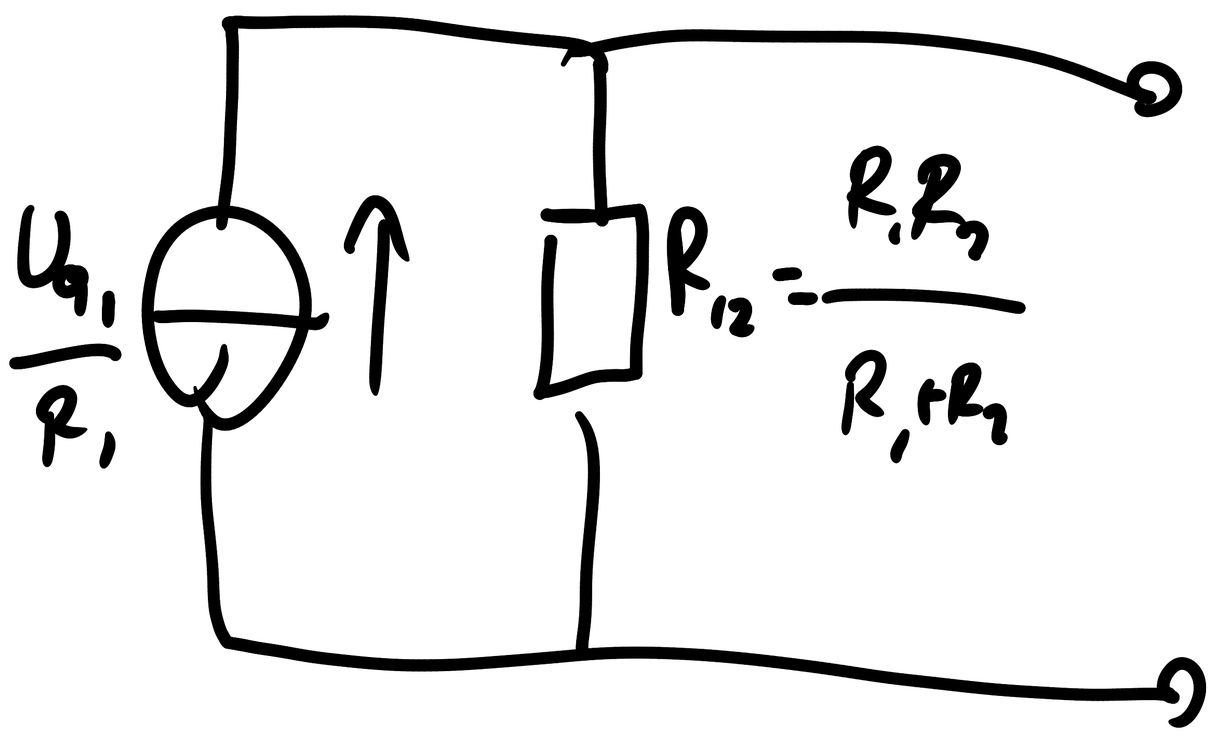
$$R_{AB} = R_1+R_2+\frac{R_1R_2}{R_3} + \frac{\left(R_2+R_3+\frac{R_2\cdot R_3}{R_1}+R_5\right)\left(R_3+R_1+\frac{R_3\cdot R_1}{R_2}+R_4\right)}{R_2+R_3+\frac{R_2R_3}{R_1}+R_5+R_3+R_1+\frac{R_1R_3}{R_2}+R_4}$$



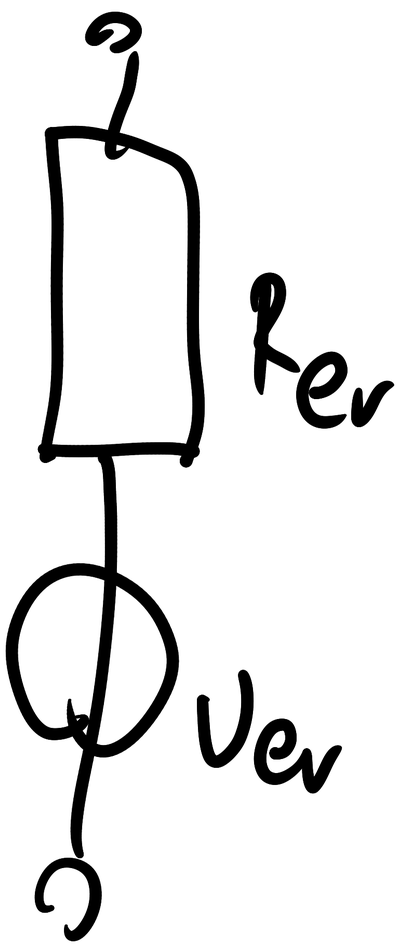
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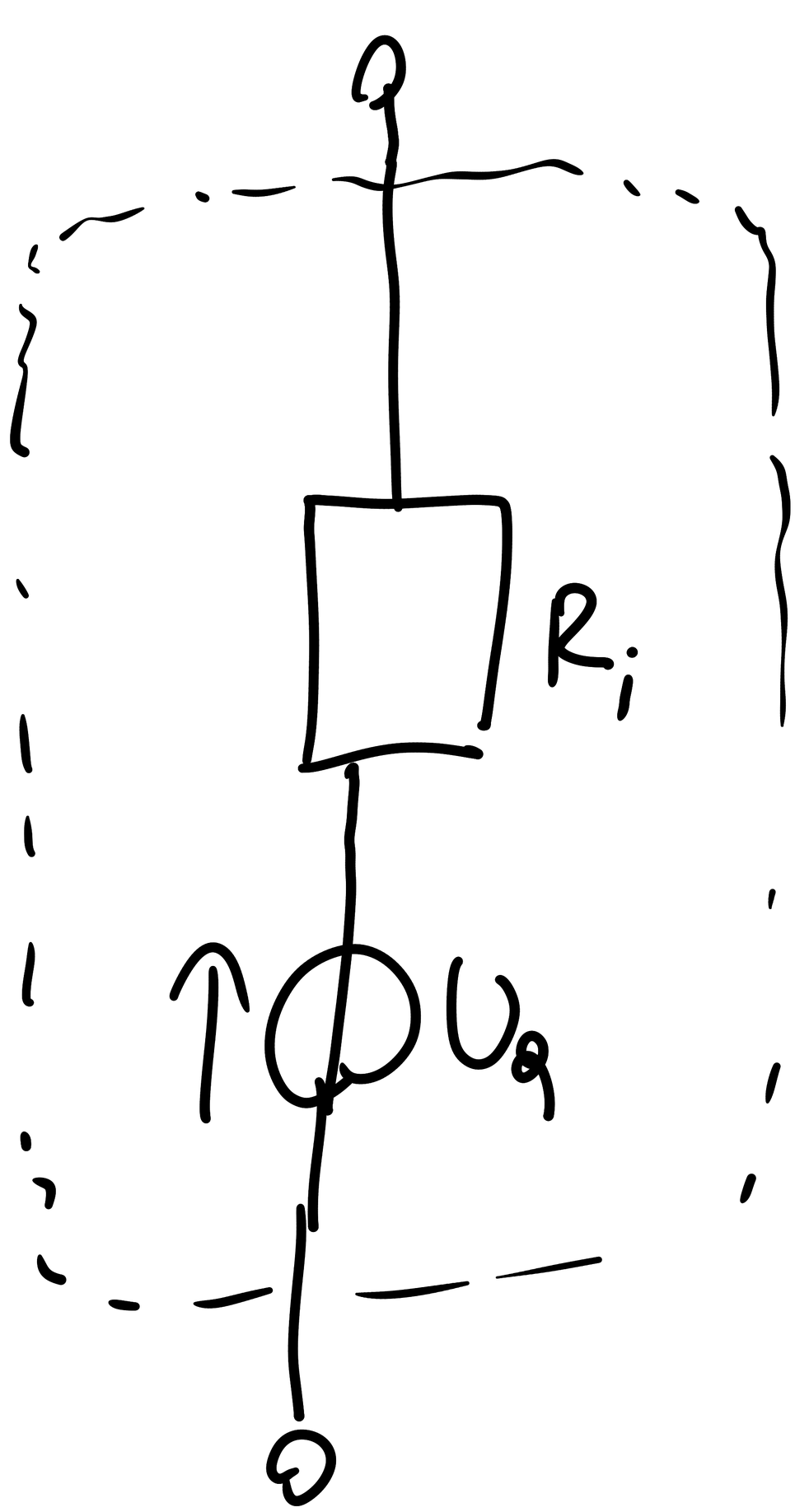
$$I_{g1} = \frac{U_{g1}}{R_1}$$

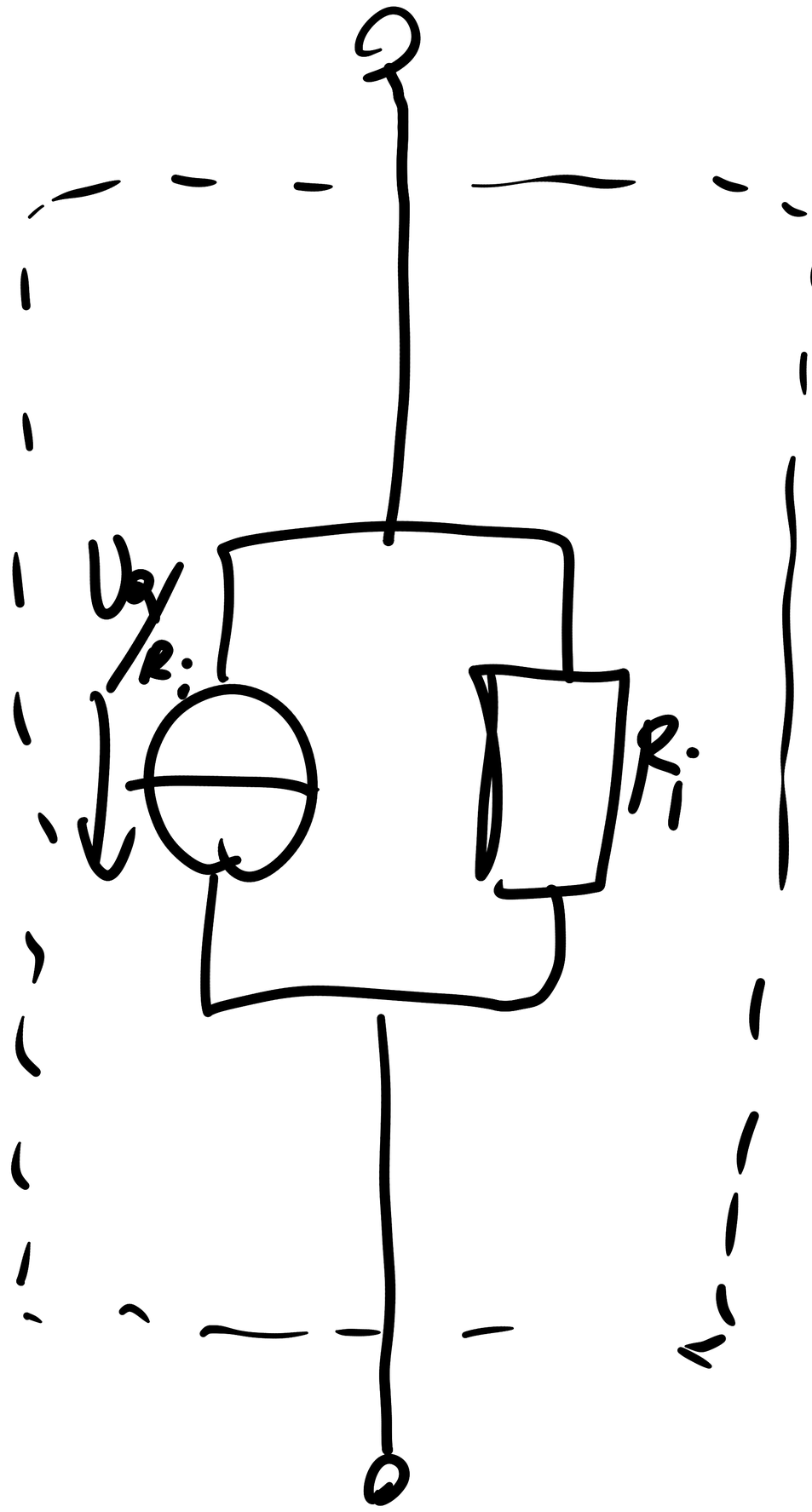


≡



$$U_{ev} = \frac{U_{g1}}{R_1} \cdot \frac{R_1 R_2}{R_1 + R_2} = U_{g1} \frac{R_2}{R_1 + R_2} \qquad R_{ev} = \frac{R_1 R_2}{R_1 + R_2}$$


R_i
U_q


U_q
R_i
R_i

